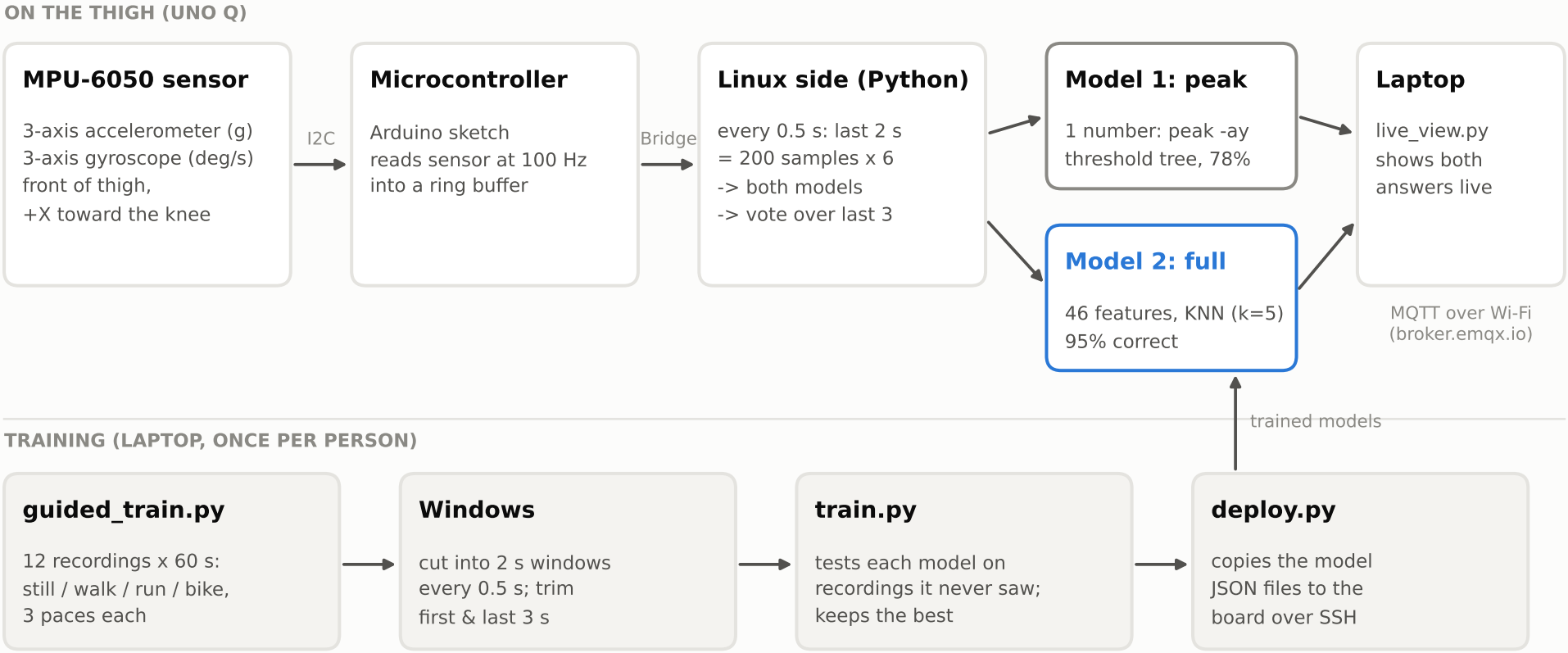
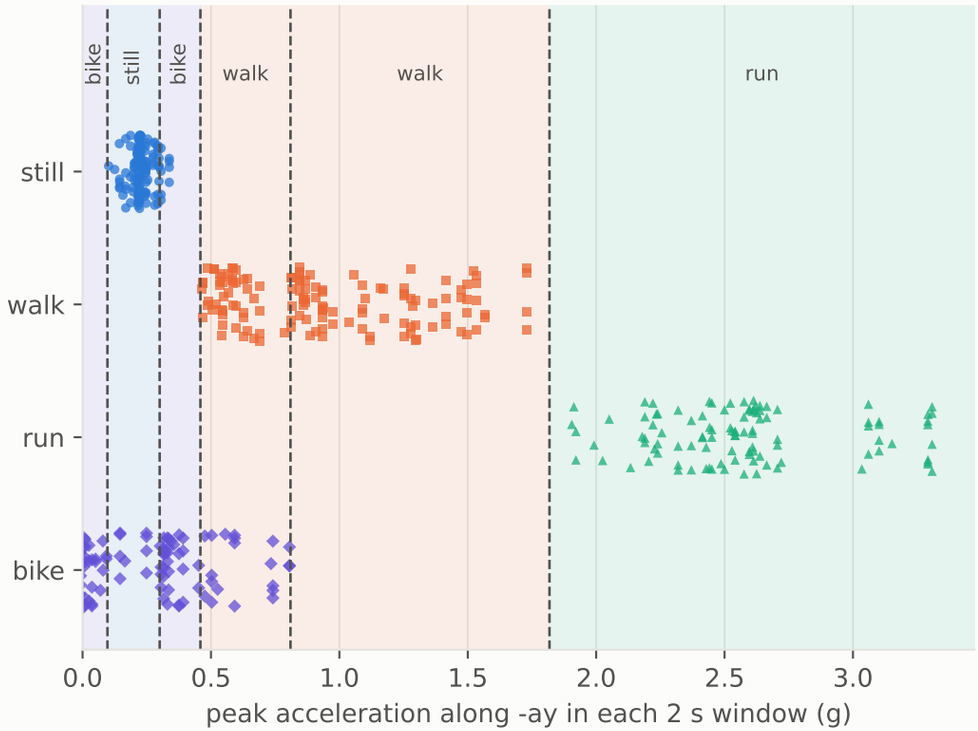
Gait identifier: how it works

The board classifies on its own every 0.5 s; the laptop only trains the models and displays the answers.



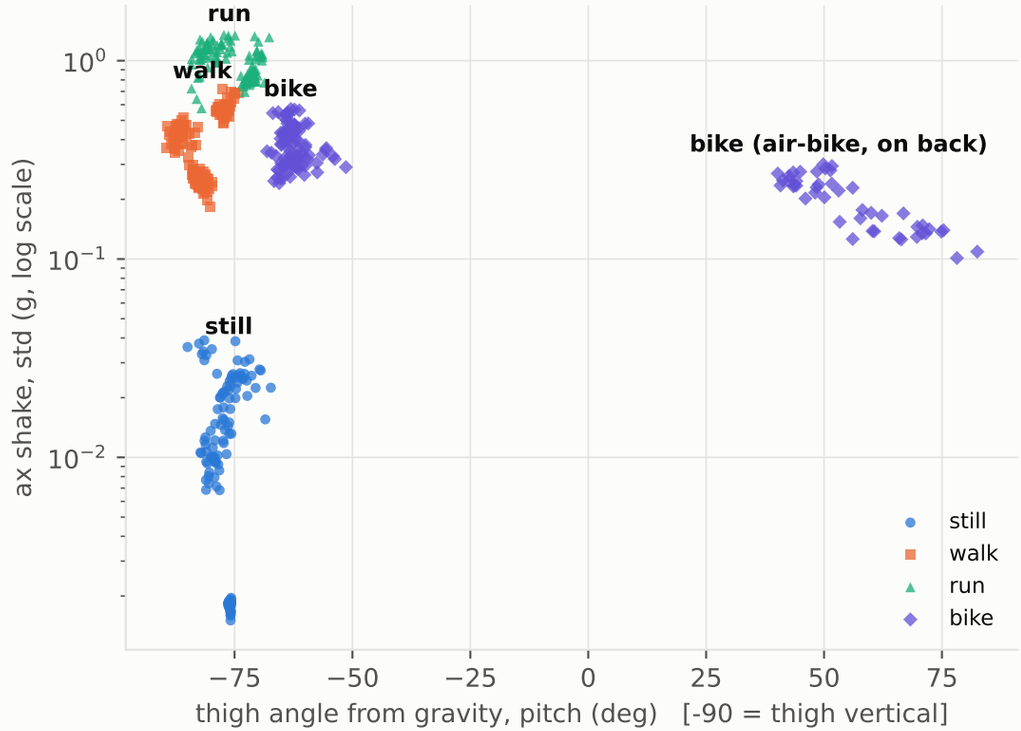
Model 1 vs Model 2 (fiona's recordings)

A. Model 1 sorts windows by ONE number



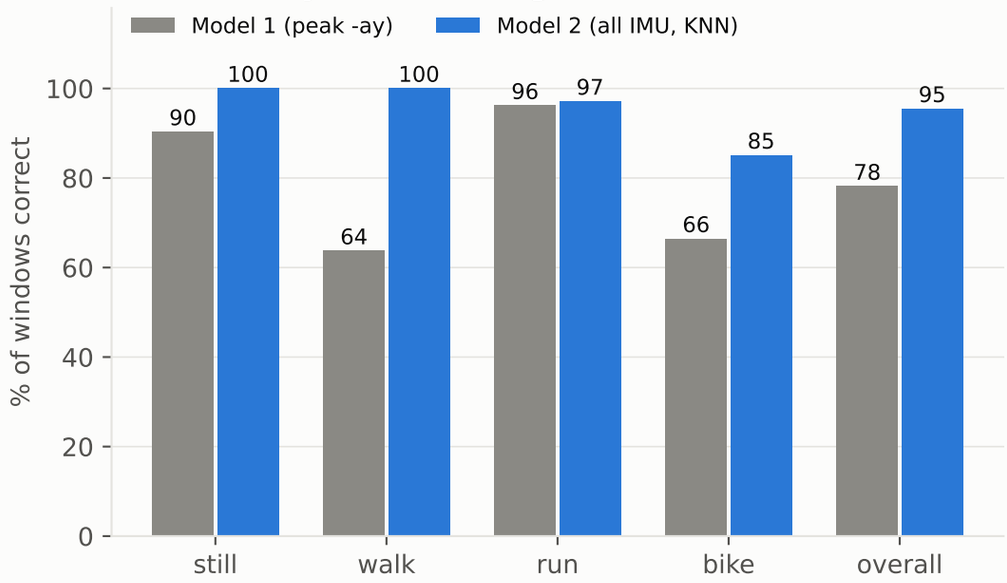
Shaded bands = the activity the learned thresholds (dashed) assign. Walk and bike overlap, so no threshold can separate them.

B. Model 2 sees many features at once



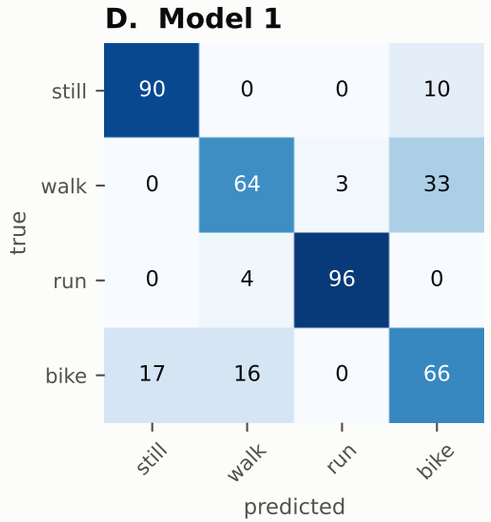
Just 2 of its 46 features: shake separates still from moving; thigh angle separates bike (thigh tilted forward) from walk and run (thigh near vertical).

C. Accuracy on recordings the model never saw

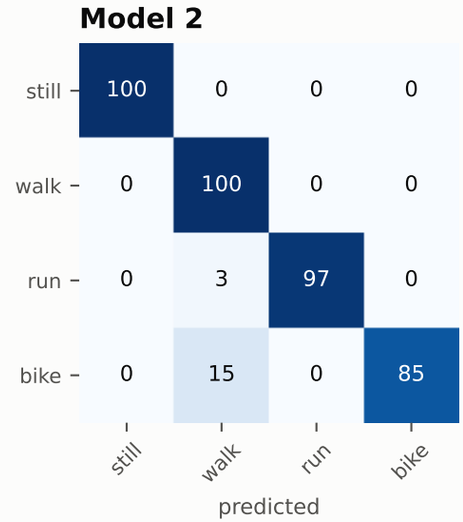


Leave-one-recording-out: train on 11 recordings, test on the 12th, repeat for all 12.

D. Model 1



Model 2



Each row: where that activity's windows went (%). Model 1 confuses walk and bike; Model 2's only real mistake is some bike windows called walk.